Pedagogical Architecture and Exercise Design for Relative Pitch and Audiation in Digital Ear Training

Diagnostic Audit of Existing Curricula and the Audiation Paradigm

Contemporary digital ear training applications frequently replicate a structural flaw inherited from traditional conservatory curricula by treating aural skills as a collection of bottom-up, atomistic discrimination tasks rather than a unified, top-down cognitive system[1]. An examination of the existing application structure reveals a strong foundation in functional scale-degree recognition alongside an over-allocation of modular space to isolated interval mechanics and static vertical chord classification[3]. Specifically, while the existing scale-degree ladder mirrors Alain Benbassat's *Functional Ear Training* progression—advancing from a single tetrachord in C major to full-octave, multi-octave, non-C static keys, and finally randomized keys with cadential grounding—the broader curriculum still devotes 36 modules to interval identification and comparison, and 40 modules to vertical chord recognition and inversions[4].

In *Aural Skills Acquisition*, Gary Karpinski demonstrates that empirical music cognition research shows minimal correlation between a learner's ability to identify isolated intervals and their capacity to comprehend or transcribe functional music in real time[5]. When learners rely on successive interval identification by measuring note-to-note distances, they treat every incoming pitch as an arbitrary new reference point[3]. This serial calculation induces severe cognitive bottlenecks in working memory and causes cumulative pitch drift during sight-singing or dictation[3]. Conversely, expert relative pitch operates from the top down: the listener first establishes the overarching tonality (resting tone) and meter, maps the primary harmonic functions (I, IV, V), locks onto stable guidepost scale degrees ($\hat{1}, \hat{3}, \hat{5}$), and processes intervals merely as secondary byproducts of scale-degree gravity[3].

Furthermore, traditional software drills primarily train *aural perception*—the immediate sensory reaction to physically sounding frequencies—rather than *audiation*[9]. Coined by Dr. Edwin Gordon within Music Learning Theory (MLT), audiation is the cognitive process of hearing and comprehending music silently when the physical sound is no longer present or has never been physically performed[9]. Just as thought is to language, audiation is to music[9]. A user who mimics a melody immediately after playback is often engaging only in sensory imitation via short-term echoic memory without syntactic comprehension[9]. True audiation requires the mind to organize sounds contextually by identifying the resting tone (Do in major, La in minor), feeling the macrobeat and microbeat subdivisions, and predicting implied harmonic functions and scale-degree resolutions before producing sound[7].

To transition the existing application from a testing-oriented perception tool into a

comprehensive audiation training environment, the existing nine activities require targeted algorithmic and pedagogical modifications alongside entirely new cognitive modules[1].

Existing Activity Module	Current Structural Implementation	Identified Pedagogical Limitation	High-Leverage Architectural Upgrade
Scale Degrees	C bottom/top half $\rightarrow$ 1 octave $\rightarrow$ multi-octave $\rightarrow$ non-C $\rightarrow$ random keys with cadence (Major/Minor/Chromatics)[4].	Users can answer via timbre memory in static-key levels or guess labels without internalizing the resolution pathway[4].	Enforce a post-answer stepwise resolution animation and audio trace (Fa-Mi-Re-Do down, Sol-La-Ti-Do up) after every response, and insert a mandatory 1-beat audiation pause before revealing answer tiles[4].
Interval Identification	16 EarMaster-style modules: ascending, descending, harmonic, mixed (introduced harmonically first)[14].	Encourages acontextual, note-to-note distance calculation disconnected from tonal gravity[3].	Reframe intervals as scale-degree pairs within an active key drone, or require the user to sing the intervening scale steps (stepwise fill-in) to verify interval size[3].
Interval Comparison	5 pools $\times$ 4 relations (shared start, shared note, nearby, unshared) = 20 modules[14].	Purely psychoacoustic pitch-distance discrimination; lacks syntactic musical function[3].	Compress into an adaptive warm-up calibration drill rather than a 20-module core track, freeing curriculum space for tonal pattern comparison[9].

Interval Singing	Up/down from Do, then each degree above/below Do (complementary intervals)[3].	Users often lunge vocally at wide or descending intervals without an internal reference anchor[12].	Explicitly scaffold Bruce Arnold's *Additive* and *Complementary* 2-step vocal chains (e.g., sing P5 $\rightarrow$ add M2 for M6; sing ascending m3 $\rightarrow$ drop octave for descending M6)[17].
Chord Identification	22 vertical recognition modules[14].	Trains vertical sonority timbre rather than linear harmonic function[8].	Pair vertical playback with mandatory voice-leading resolution (e.g., resolving V^7 tendency tones Ti $\rightarrow$ Do and Fa $\rightarrow$ Mi)[3].
Chord Inversions	18 vertical inversion modules[14].	Students struggle to isolate the bass note from upper-structure voicings[18].	Require the user to sing the bass note first, then arpeggiate the chord upward in solfege (Mi-So-Do for I^6) to lock in inversion geometry[19].
Scale Identification	13 lessons across 14 scales[4].	Passive label matching on linear scalar runs rarely transfers to real melodies[2].	Present scales as randomized, non-scalar key profile melodies (Temperley & Marvin model) or short characteristic modal grooves, requiring

			resting-tone identification[7].
Melodic Dictation & Sight Singing	Do and Re first, adding one degree per lesson; 4/4 and 3/4 with 8th notes; minor/other keys[3].	Jumps directly from stimulus to staff notation, bypassing short-term memory chunking and protonotation[18].	Insert Karpinski's *Protonotation* phase (mapping pulse and contour without staff lines) and Kodály *Silent Singing* drop-out gates during sight-singing[2].
Intervals in a Key	Diatonic and chromatic intervals within an established key[3].	Static pair identification lacks phrase-level rhythmic and harmonic momentum[3].	Transform into Jersild-style *Tendency Tone Pair* and *Tonal Pattern* drills focusing on active resolution (So $\rightarrow$ Do, Ti $\rightarrow$ Do, Fa $\rightarrow$ Mi, Fi $\rightarrow$ So)[3].

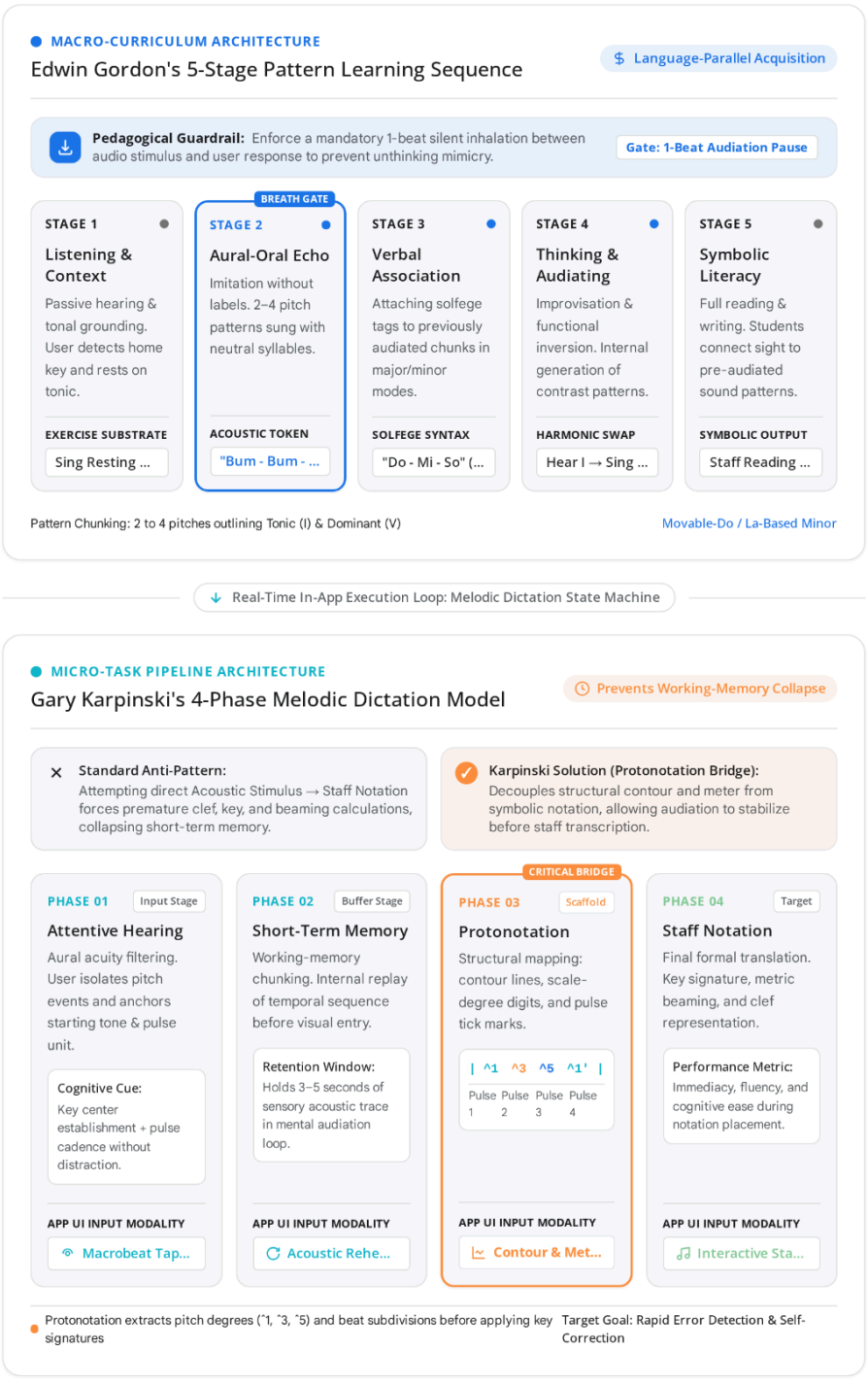
Edwin Gordon's Music Learning Theory (MLT) and Tonal Pattern Mechanics

A foundational upgrade for the application involves integrating Edwin Gordon's Music Learning Theory (MLT), which models music acquisition on native language acquisition[20]. In language development, infants spend months absorbing a listening vocabulary of individual words before speaking, thinking in sentences, reading, and writing[20]. In music, songs are equivalent to sentences, whereas **tonal patterns** and **rhythm patterns** are equivalent to individual words[13]. Gordon's research demonstrates that listening to whole songs does not automatically build a functional listening vocabulary; rather, learners must internalize a taxonomy of short tonal patterns consisting of 2 to 5 pitches outlining tonic, dominant, and subdominant functions divorced from rhythmic complexity[20].

Crucially, MLT dictates that **context** (tonality and meter) must always precede **content** (specific patterns), and that humans learn what a musical structure *is* primarily by contrasting it with what it *is not*[7]. Therefore, rather than restricting early modules exclusively to Major and Minor, the application should expose learners across eight contrasting tonalities—Major (Do resting tone), Harmonic Minor (La resting tone with raised Si), Dorian (Re), Phrygian (Mi), Lydian (Fa), Mixolydian (So), Aeolian (La with natural So), and Locrian (Ti)—while anchoring primary

pattern drills in Major and Minor[4]. When bridging these patterns to melodic dictation, the software architecture should pair Gordon's macro-curriculum with Gary Karpinski's four-phase cognitive dictation pipeline—progressing systematically through hearing, short-term memory chunking, structural understanding via protonotation, and finally symbolic notation—to prevent working-memory collapse[13].

Bridging Acoustic Perception and Symbolic Literacy Through **Sequential Audiation**



The upper tier maps Edwin Gordon's five-stage Music Learning Theory (MLT) sequence into discrete software modules, enforcing an audiation breath pause between stimulus and vocalization. The lower tier details Gary Karpinski's four-phase melodic dictation model, highlighting how extractive listening and protonotation prevent working-memory collapse prior to staff notation.

Data sources: TopMusic Arts, The Improving Musician (MLT for Choirs), Texas State University (Karpinski Dictation Research), Journal of Music Theory Pedagogy, The Improving Musician (Gordon MLT Guide), Everyday Musicality

To operationalize MLT within the software architecture, four specific exercise mechanics should be integrated prior to or alongside the existing Melodic Dictation and Sight Singing modules[13].

Resting Tone Drop-Needle ("Find Home")

Developing an unshakeable sense of the resting tone is the foundational prerequisite of audiation[7]. In this activity, the application plays an unfamiliar acoustic or MIDI melody, or a contrapuntal excerpt, that clearly outlines a tonality without relying on a stereotypical $I - IV - V - I$ block cadence[20]. At a randomized point mid-phrase—or immediately prior to the final cadential resolution—playback halts abruptly[20]. A visual prompt cues the user to take a deep breath lasting one macrobeat and sing the resting tone into the microphone[20]. At the initial *Aural/Oral* level, the user sings the resting pitch on a neutral syllable ("bum") so they internalize the acoustic sensation without cognitive label interference[13]. At the subsequent *Verbal Association* level, the user must both sing the correct resting pitch and tap its functional solfege syllable (Do for Major, La for Minor, Re for Dorian), thereby identifying and labeling the overarching tonality entirely by ear[7].

Bimodal Tonal Pattern Echo (The "Audiation Breath" Protocol)

Tonal pattern drills present arpeggiated and stepwise pitch groupings of 2 to 5 notes—such as Major Tonic (Do-Mi-So, Mi-So-Do, So-Mi-Do), Major Dominant (So-Ti-Re, So-Fa-Re-Ti), Minor Tonic (La-Do-Mi), and Minor Dominant (Mi-Si-Ti, Re-Ti-Si)—completely divorced from objective rhythm so that every pitch sounds as an even, unmeasured pulse[26]. Two critical software mechanics govern this exercise to ensure the learner is audiating rather than merely mimicking[13]. First, the application must implement a mandatory breath gate: when students echo a prompt immediately, they rely on short-term acoustic imitation and forget the pattern rapidly[20]. By enforcing a 1-beat silent delay between the end of the application's prompt and the opening of the microphone recording window—accompanied by a visual "Breathe & Audiate" expansion ring—the interface forces the user to audiate the tonality and pitch relationships during the physical act of inhalation[13]. Second, following Gordon's Learning Sequence Activities, each pattern is executed across two distinct passes: a *Teaching Mode* pass and an *Evaluation Mode* pass[13]. On the first pass (*Teaching Mode*), the application sounds a subtle tonic reference or harmonizes the initial pitch of the user's echo to build confidence and establish pitch center[13]. On the second pass (*Evaluation Mode*), the user must perform the pattern completely unaccompanied[13]. Across curriculum levels, this exercise progresses strictly from Neutral Syllable echoes ("bum") to Solfege echoes (Do-Mi-So), and finally to Syllable Translation, wherein the app sings a pattern on "bum" and the user echoes it back translated into functional solfege[13].

Functional Conversational Improvisation ("I–V Dialogue")

Gordon's aptitude research reveals a counterintuitive cognitive phenomenon: regardless of musical aptitude, it is more difficult for a learner to accurately repeat an exact tonal pattern just performed by someone else than to perform a contrasting one, yet generating a functional response is precisely what transitions a learner from imitation to true assimilation and generalization[9]. In the *I–V Dialogue* module, the application initiates a musical conversation by

singing a 3- or 4-note pattern, and the user must improvise a vocal response into the microphone under progressive harmonic constraints[13]. In the initial pattern-differentiation stage, the app sings a Tonic pattern (Do-Mi-So), and the user is required to sing any *different* Tonic pattern from their learned vocabulary, such as So-Do-Mi or Mi-Do-So[26]. In the intermediate harmonic-function inversion stage, the user must answer with the *opposite* harmonic function: if the app sings a Tonic pattern (Do-Mi-So), the user must immediately respond with a Dominant pattern (So-Fa-Re-Ti), whereas if the app sings Dominant, the user resolves the conversation with a Tonic pattern[26]. In the advanced first-pitch extraction stage, the app sings a rapid 4-note pattern on a neutral syllable and requires the user to isolate and sing back *only the first pitch* of the pattern in solfege[13]. This third variation directly trains the exact cognitive subskill that Karpinski identifies as the single greatest point of failure in collegiate melodic dictation: accurately determining the starting pitch of an exercise before attempting to track subsequent notes[12].

Tonal Pattern Permutation and Symbolic Generalization

Before introducing full melodic sight-singing, the application should bridge aural vocabulary to notation using Gordon's *Symbolic Association* and *Composite Synthesis* mechanisms[20]. Initially, in an aural permutation drill, the app sounds a 3- or 4-note pattern and instructs the user to sing it back in retrograde (reversed), or to sing the pattern while silently omitting a designated pitch, such as replacing the second note with an audiated rest[20]. Subsequently, the screen displays four short notated pattern cards with a visual arrow pointing to Do or La on the staff to establish keyality[26]. The user drags and drops these visual pattern cards to match the sequence of a heard folk melody, and then rearranges those same visual cards into a novel order to compose and sing a new melodic sentence aloud[20]. Crucially, when reading full melodies in this module, the software enforces the *Jump Right In* pedagogical directive: users are instructed to audiate the melody silently—singing only discrete tonal patterns aloud with tonal syllables if needed—and then perform the complete melody on a neutral syllable or instrument rather than mechanically decoding the entire melody note-by-note with solfege[33].

Cognitive Subskills, Working Memory, and Kodály Inner Hearing

A persistent contradiction in traditional ear training is that exercises such as melodic dictation and sight-singing *test* pitch internalization without directly *teaching* the cognitive mechanisms required to achieve it[12]. As Edward Klonoski (1998), Sarah Gates (2021), and Timothy Chenette (2021) demonstrate, learners who struggle with relative pitch rarely suffer from perceptual deafness; rather, they exhibit deficits in three foundational subskills: subvocal motor mapping (connecting mental imagery to the physical sensations of the vocal folds, lips, and tongue), attentional control (filtering target pitches from sonic interference), and working memory capacity (holding and manipulating pitch chunks across time)[1].

Combining cognitive psychology paradigms with Zoltán Kodály's *inner hearing* ("silent singing") pedagogy yields five highly structured exercises that directly train these underlying mechanisms by manipulating the temporal gap between acoustic stimulus and vocal output[2].

Temporal Mechanics of **Inner Hearing** and Working Memory Drills

Beat-by-beat signal architecture comparing acoustic stimuli, internal audiation states, and evaluation gates.

SIGNAL STATE KEY: App Audio (Acoustic) · Silent Audiation (Internal) · Mic Evaluation Gate · Mental Rule / Swap

1 Kodály Silent Singing Drop-Out Gate

4/4 Metre • Internal Pitch & Pulse Preservation

	Measure 1: Audio Anchor	Measures 2–3: Muted Gate (Silent Audiation)	Measure 4: Pitch Verification
App Output	Do · Re · Mi · Fa	*Acoustic Output Muted (8 Beats Silent Internal Count)*	Metronome Click
Cognitive State	Singing Aloud With Audio	Audiating Internally: `[ Sol – La – Ti – Do5 – Ti – La – Sol – Fa ]`	✓ Target Check: Do

Evaluation Window: App opens microphone gated buffer at Measure 4 Beat 1; expects tonic vocalization within ±25 cents.

2 Musical 2-Back Canon Stream

Dual-Task Encoding • Real-Time Imitation at N=2

	Beat 1	Beat 2	Beat 3	Beat 4	Beat 5	Beat 6
App Stimulus	Do4 Stimulus 1	Re4 Stimulus 2	Mi4 Stimulus 3	Fa4 Stimulus 4	Sol4 Stimulus 5	La4 Stimulus 6
2-Back Lag						
Vocal Production	[Encoding 1]	[Encoding 2]	Do4 ✓ Sings (N-2)	Re4 ✓ Sings (N-2)	Mi4 ✓ Sings (N-2)	Fa4 ✓ Sings (N-2)

Cognitive Tax: User must listen to incoming pitch while simultaneously singing the pitch heard two beats earlier without vocal interference.

3 Mental Pitch Swap (Melodic Word-Updating)

Working Memory Transformation • In-Head Mutation

PHASE A: STIMULUS — App Audio

Do · Re · Mi · Do

Original diatonic 4-note motif presented to learner at 80 BPM.

PHASE B: MENTAL RULE — Swap Rule

Degree ^3 (Mi) → Degree ^4 (Fa)

Inner Ear: `Do – Re – Fa – Do`

PHASE C: MIC GATE — Pitch Detection

Do · Re · Fa (Target ✓) · Do

App validates pitch accuracy with strict weight on swapped degree (^4).

4 Additive & Complementary Interval Scaffolding

Cognitive Rule of 9: Descending M6 = Ascending m3 – Octave

MENTAL PITCH AXIS

- Eb4 (+3 Semitones) — Ascending m3 Guide
- `↓ Octave Inversion Jump (-12 st)`
- C4 (Tonic Root: 0 st) — App Anchor Pitch
- `↓ Descending Target (-9 st)`
- Eb3 (-9 Semitones) — Target: Desc M6 ✓

1. **Sound Reference Tonic (C4)**
   App plays root. Learner anchors tonal center in working memory.
2. **Audiate Complementary Minor 3rd (Eb4)**
   Easier ascending interval (`+3 st`) is audiated, then dropped 1 octave down (`-12 st`).
3. **Sing Resulting Descending M6 (Eb3)**
   User vocalizes final pitch directly below tonic. Mic validates pitch landing within gate.

Each exercise manipulates the temporal gap between acoustic stimulus and vocal production to force genuine audiation. In the Kodaly Drop-Out Gate, the user maintains internal pitch across muted measures; in the Musical 2-Back and Mental Swap drills, working memory is taxed via concurrent encoding and silent pitch substitution.

Data sources: Journal of Music Theory Pedagogy (Chenette), Journal of Music Theory Pedagogy (Internalization), Informit, DSMusic, Guitar Lessons Myrtle Beach

Kodály Audiation Gate ("Silent Singing" Drop-Out)

In the Kodály tradition, inner hearing is verified by having students alternate between singing aloud and singing silently inside their heads at a conductor's signal while maintaining a steady physical pulse[2]. Within the application's Sight Singing and Scale Degree modules, this translates into an **Audiation Drop-Out Gate** that operates across phrase-level, metric, and flash-memory dimensions[2]. In the phrase-level drop-out mode, the user sight-sings a 4- or 8-measure melody accompanied by a visual metronome pulse, singing Measure 1 aloud into the microphone[2]. Across Measures 2 and 3, a red "Silent Singing" gate activates, requiring the user to cease vocalizing aloud and trace the notated pitches purely through subvocal inner hearing—initiating the physical vocal-tract posture of each pitch without phonating—while continuing to tap the macrobeat on the screen[2]. On Beat 1 of Measure 4, a green "Sing Aloud" gate triggers, and the DSP engine evaluates whether the user's re-entry pitch matches the exact target scale degree without having drifted toward their habitual vocal "comfort pitch"[2]. For an advanced rhythmic-tonal integration drill, the metric drop-out mode instructs the user to perform a known or notated melody in $2/4$ or $3/4$ while silently inner-hearing every note that falls on the strong beat (Beat 1 of each bar) and singing only the weak beats aloud; on the immediate repetition, the user reverses the polarity, singing only Beat 1 aloud and inner-hearing all weak beats[24]. Complementing these real-time drop-outs, a flash-score recall variant emulates Kodály's *333 Elementary Exercises* memory drills by displaying a 2-measure score on screen for 10 to 15 seconds while sounding only the tonic pitch[2]. The user must silently read and audiate the score before the notation vanishes, after which they sing or notate the phrase entirely from internal rehearsal[2].

Melodic Word-Updating (Mental Pitch Replacement)

Adapted by Timothy Chenette from cognitive psychology working-memory paradigms, the **Melodic Updating** task trains the ability to hold an aural image in working memory and actively manipulate its components without external sound support[6]. In this drill, the application plays a 4- to 6-note diatonic melody, such as Do – Re – Mi – Do – Mi – Re, without displaying staff notation[6]. Immediately following playback, a text and audio prompt instructs the user to perform a surgical mental substitution—for example, silently replacing every instance of Mi with Fa, replacing the third pitch with So, or shifting the entire melodic contour up one diatonic step to begin on Re[6]. Following a 3-second silent processing window during which the user rehearses the altered sequence with their inner voice, the user sings the newly mutated melody (Do – Re – Fa – Do – Fa – Re) into the microphone[6]. Over progressive difficulty tiers, two or three sequential pitch swaps are chained together on the same base melody before vocalization, building exceptional stability in mental pitch representation[6].

Musical N-Back and Delayed Canon (Dux/Comes)

In cognitive science, the N-back task is a primary vehicle for expanding working memory capacity and attentional control[6]. Translated into an aural skills module, the **Musical N-Back** operates through both recognition and vocal production modes[6]. In the recognition mode, the application plays a continuous, slow stream of single diatonic pitches—sounding one note

every 2 seconds over a subtle tonic drone—and the user taps a "Match" button whenever the currently sounding scale degree is identical to the pitch heard exactly N steps earlier in the series, progressing from $N = 2$ to $N = 3$ [6]. Because no notation or solfege labels are required during playback, this mode isolates pure tonal working memory and active encoding[6].
In the vocal canon mode, which models historical counterpoint improvisation and Kodály hand-sign canons, the application plays a slow, continuous melodic line acting as the leading voice (*dux*)[2]. The user acts as the following voice (*comes*), singing the exact pitch that the application sounded N notes ago (1-back or 2-back) at the precise moment the application sounds the next new pitch[6]. This dual-task architecture forces the user's auditory cortex to simultaneously encode an incoming external pitch while executing vocal motor output for a previously stored pitch[6].

Extractive Listening and the Protonotation Canvas

Standard melodic dictation modules frequently overwhelm beginners because they conflate four distinct cognitive phases identified by Karpinski: **Hearing** (attentive perception), **Short-Term Melodic Memory** (extractive listening and chunking), **Understanding** (metric and scale-degree inference), and **Notation** (translating sound into staff symbols)[6]. When a user fails a dictation in an application, the failure is frequently caused by a bottleneck in memory capacity or notation speed rather than a deficit in pitch perception[6].
To isolate and train short-term memory amidst realistic musical textures, the application should incorporate **Extractive Listening Drills** wherein a full 4-measure excerpt or polyphonic two-voice passage is played, and the user is instructed to selectively memorize *only* the first two measures, *only* the cadential arrival, or *only* the lower contrapuntal voice despite sonic interference from the surrounding music[6]. Furthermore, before unlocking 5-line staff dictation, users should complete dictations on an interactive **Protonotation Canvas**[18]. While listening to the melody, the user taps the screen to lay down vertical macrobeat and microbeat lines—establishing duple or triple meter visually without time-signature arithmetic—and sketches continuous pitch contours or drags solfege/scale-degree tokens onto the beat grid[6]. Only after the user verifies their audiated understanding via protonotation and vocal sing-back does the interface transition them to translating those musical chunks onto a formal clef and key signature[18].

Additive and Complementary Interval Scaffolding

To upgrade the existing Interval Singing and Interval Identification modules, the application should embed the **Additive** and **Complementary** techniques codified by Bruce Arnold, Alain Benbassat, and Keith Wyatt & Carl Schroeder[17]. Rather than asking a user to sing or identify a difficult interval—such as an ascending minor 7th or a descending Major 6th—in a single unguided leap, the user interface provides a two-step scaffolding protocol[17]. Under the additive approach, the user first sings a familiar anchor interval, such as a Perfect 5th or Perfect Octave, and then adds or subtracts a major or minor second or third to reach the target pitch[17]. For instance, when prompted for an ascending Major 6th, the app guides the user to hum a Perfect

5th (Do $\rightarrow$ So) and then step up a Major 2nd (So $\rightarrow$ La); to find a Major 7th, the user audiates a Perfect Octave and steps down a half-step (Do $\rightarrow$ Ti)[17].

Under the complementary approach, which addresses the acute difficulty of audiating wide descending intervals directly from an upper reference note, the software trains the rule of interval inversion, wherein ascending and descending interval numbers sum to nine and qualities invert (Major becomes minor, minor becomes Major, and Perfect remains Perfect)[17]. When prompted to sing a descending minor 7th from a given pitch, the user is guided to first audiate or quietly hum an *ascending Major 2nd* from that reference pitch, lock onto that pitch class in working memory, and displace it down one octave[17]. Similarly, a descending Major 6th is executed by audiating an ascending minor 3rd and dropping it an octave, while a descending Perfect 4th is located by audiating an ascending Perfect 5th and inverting it downward[17].

Linear Harmony, Guide-Tone Voice Leading, and Non-Chord Tone Resolution

A major gap in the existing curriculum is the bridge between isolated vertical chords—represented by the 22 Chord Identification and 18 Chord Inversion modules—and linear harmonic progressions in real music[8]. As Gordon emphasizes, teaching the vertical voicing of a single chord pales in comparison to training students to audiate linear chord changes, because harmonic audiation is fundamentally horizontal[8]. Functional harmonic hearing relies on tracking the bass line (harmonic roots), following the smooth voice-leading of interior guide tones (3rds and 7ths), and separating structural chord tones from melodic embellishments (non-chord tones)[7].

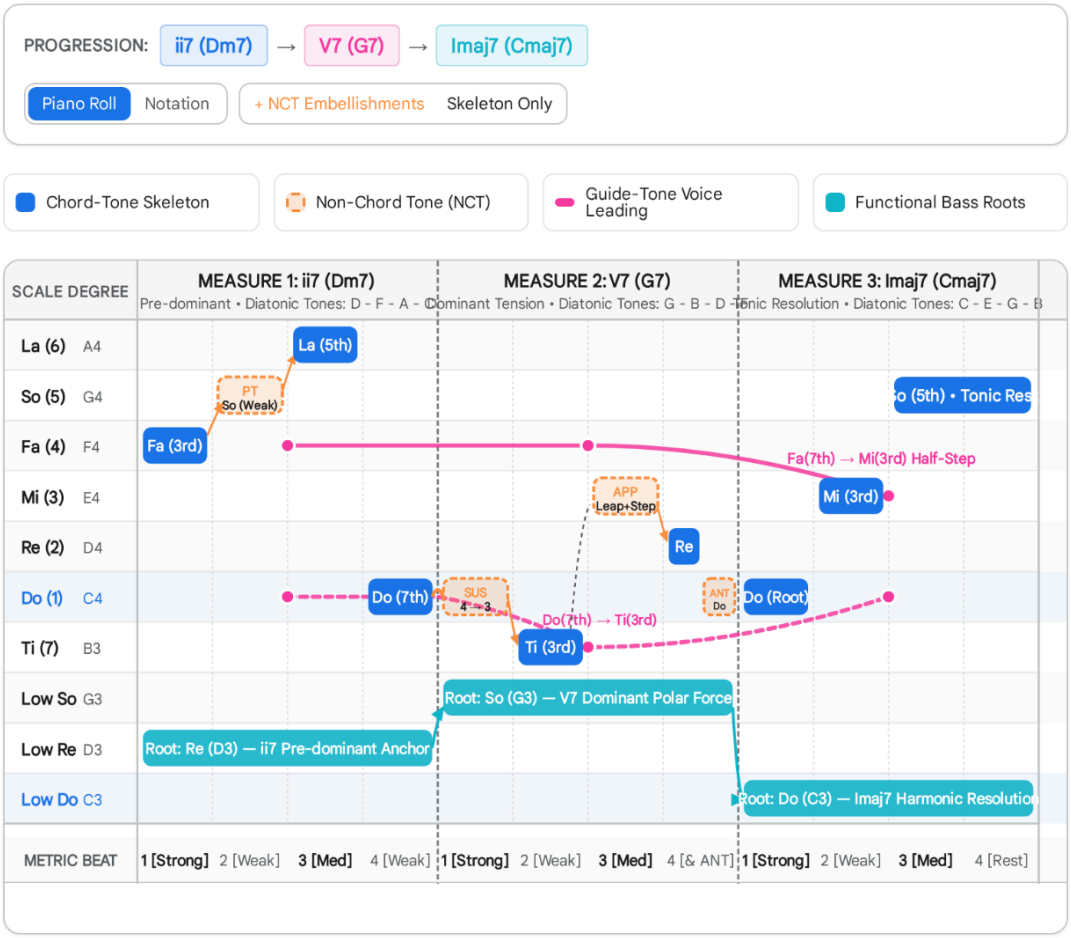
Linear Bass-Line and Guide-Tone Singing

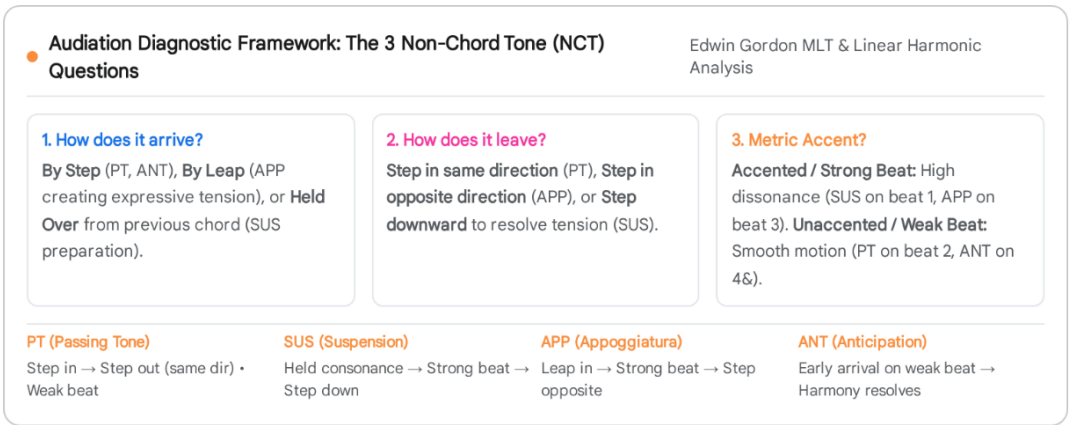
Instead of presenting chord progressions solely as multiple-choice Roman numeral quizzes, the application should introduce an interactive **Harmonic Voice-Leading Vocalizer** that advances across three linear stages[19]. In the initial chord-root bass tracking stage, the application plays a harmonized melody built on primary I, IV, V progressions in Major and Minor, as well as I, IV subdominant progressions in Dorian and Mixolydian[7]. On the first pass, the user sings the root of each chord (Do – Fa – So – Do) in unison with the bass line; on the second pass, the app plays *only* the upper melody while the user sings the implied bass roots into the microphone, or sings broken tonic and dominant triads in a rhythmic pattern against the melody[19].

In the intermediate single-reference guide-tone stage, adapted from jazz ear-training pedagogy, the exercise strips away instrumental accompaniment to force deep internal harmonic projection[38]. The screen displays a harmonic progression—such as $ii^7 - V^7 - I^{\Delta 7}$ in Major or $ii^{\varnothing 7} - V^{7(\flat 9)} - i^7$ in Minor—and sounds *only a single static reference pitch* (the tonic Do or La) prior to the exercise[38]. Without any chords sounding, the user must audiate and sing the connected voice-leading line of the **guide tones** (the 3rds and 7ths of each chord, where the 7th of one chord resolves stepwise down by half- or

whole-step to become the 3rd of the subsequent chord: e.g., Do [7th of ii^7] $\rightarrow$ Ti [3rd of V^7] $\rightarrow$ Ti/Do [7th/root of $I^{\Delta 7}$], or Fa [3rd of ii^7] $\rightarrow$ Fa [7th of V^7] $\rightarrow$ Mi [3rd of $I^{\Delta 7}$])[3]. Once the user can sing these voice-leading threads from a single reference tone in both major and minor contexts, the third stage overlays piano rootless voicings (or shell voicings with roots in the bass and 3rds and 7ths in the upper register), requiring the user to sing the roots, 3rds, 5ths, or 7ths cleanly through shifting harmonic colors[19].

Deconstructing Surface Melody into **Harmonic Skeletons** and Guide-Tone Lines





Rather than testing chords as isolated vertical blocks, the application trains linear audiation across three synchronized layers: stripping away Non-Chord Tones (orange dashed blocks) to reveal the structural Chord-Tone Skeleton (solid blue), tracking smooth half-step Guide-Tone resolutions between 3rds and 7ths (magenta), and anchoring functional Bass Roots (cyan).

Data sources: Alfred Music Blog, Gordon Institute for Music Learning, Free Jazz Lessons, r/musictheory, Dixboro Music Academy, Motifkit

Non-Chord Tone (NCT) Skeleton and Diagnostic Explorer

When learners transition from isolated scale-degree drills to transcribing real repertoire, they frequently lose their tonal bearings because real melodies rarely stick strictly to the notes of the underlying chord[39]. Karpinski notes that introducing new melodic concepts as neighboring and passing tones to the tonic triad is essential for structural hearing[40]. Following modern interactive pedagogical frameworks, the application should include a **Non-Chord Tone Explorer & Diagnostic Module** featuring a dual Notation / Piano Roll view switcher—allowing non-readers to visualize pitch height and duration geometrically—and a **Skeleton Toggle** button[39].

In practice mode, the user listens to a short harmonized phrase and toggles between the *Decorated Melody* and the *Chord-Tone Skeleton*, which strips away all non-harmonic tones so the ear perceives the underlying structural framework[39]. In assessment and improvisation modes, the user identifies, sings, or adds ten distinct non-chord tone archetypes by evaluating three diagnostic questions: how the pitch arrives (by step, by leap, or held over from the preceding chord), how it leaves (by step or leap, in the same or opposite direction), and where it lands metrically (on a strong accented beat versus a weak unaccented beat)[39].

Non-Chord Tone Archetype	Melodic Arrival Mechanism	Melodic Departure / Resolution	Metric Placement	Perceptual Character & App Exercise Mechanic
Passing Tone (PT)	Step from chord tone[39]	Step in *same* direction to chord tone[39]	Weak (unaccented)[39]	Smooth scalar motion filling a 3rd; user toggles skeleton vs. filled third (Do-[Re]-Mi)[39].
Accented Passing Tone (APT)	Step from chord tone[39]	Step in *same* direction to chord tone[39]	Strong (accented)[39]	Lands directly on the downbeat, creating acute harmonic friction before stepping into the chord tone[39].
Neighbor	Step from	Step in	Weak	Brief

Tone (NT)	chord tone[39]	*opposite* direction back to same note[39]	(unaccented)[39]	decorative upper or lower dip around a stationary chord tone (Mi-[Fa]-Mi)[39].
Suspension (SUS)	Held over (tied) from prior chord consonance[39]	Step *downward* to chord tone[39]	Strong (accented)[39]	Classic $4-$ or $7-$ tension held across a chord change and released downward (Fa $\rightarrow$ Mi over I)[39].
Retardation (RET)	Held over (tied) from prior chord consonance[39]	Step *upward* to chord tone[39]	Strong (accented)[39]	Tied dissonance resolving upward (typically Ti held over into I, resolving up to Do)[39].
Appoggiatura (APP)	*Leap* from chord tone[39]	Step in *opposite* direction to chord tone[39]	Strong (accented)[39]	Expressive, unprepared lean into a strong-beat dissonance before stepping into resolution[39].
Escape Tone / Échappée (ET)	Step from chord tone[39]	*Leap* in *opposite* direction to chord tone[39]	Weak (unaccented)[39]	Subtle detour stepping away from expected resolution before leaping to the target

				tone[39].
Anticipation (ANT)	Step (or leap) arriving *early*[39]	Remains on *same note* as harmony catches up[39]	Weak (before chord change)[39]	Rhythmic push where the melody arrives at the new chord tone a fraction of a beat ahead of the rhythm section[39].
Changing Tones / Double Neighbor (CT)	Step from chord tone[39]	Leaps to *opposite* neighbor, then steps back[39]	Usually weak[39]	Brackets a target chord tone from both above and below (Do $\rightarrow$ [Re $\rightarrow$ Ti] $\rightarrow$ Do)[39].
Pedal Point (PED)	Sustained or repeated single pitch[39]	Held stationary until harmony returns to consonance[39]	Spans multiple bars/chords[39]	Anchors a moving harmonic progression over a static tonic (Do) or dominant (So) bass drone[39].

Contextual Cadences and Secondary-Dominant Modulation Tracking

Rather than confining harmonic recognition to static four-chord loops, the application should incorporate contextual **Cadence** and **Modulation** modules modeled on Auralia's advanced syllabus and Michael Callahan's voice-leading pedagogy[14]. In the contextual cadential module, the application plays a 4-measure phrase with four-part or instrumental accompaniment and asks the user to classify the final cadence—Perfect Authentic ($V \rightarrow I$ with $\hat{1}$ in the soprano), Imperfect Authentic, Plagal ($IV \rightarrow I$), Half ($\text{any} \rightarrow V$), or Deceptive ($V \rightarrow vi$)—both with and without a preceding lead-in melody[14]. In the modulation tracking module, the app presents two-part or four-part phrases that pivot to a new key, such as tonicizing the dominant, relative minor, or subdominant[14]. This activity targets two connected learning outcomes: aurally detecting the exact chromatic alteration that implies a secondary

dominant (such as Fi acting as the raised fourth/new leading tone in V/V, or Te acting as the lowered seventh in V^7/IV), and vocally assimilating the voice-leading tendency of that altered pitch (Fi $\rightarrow$ So) before singing the new resting tone (Do) of the destination key[14].

Multi-Modal Error Detection Mechanics

Although Karpinski and Benward identify **error detection**—the ability to recognize discrepancies between written notation and a live or recorded performance—as an indispensable skill for every practicing musician, conductor, and audio producer, national curriculum surveys reveal that 84% of collegiate programs rank it as the least practiced activity in freshman aural skills[1]. Error detection is a distinct cognitive skill that does not automatically develop from dictation alone; it requires rapid multi-sensory comparison between an internally audiated score ("the hearing eye") and an incoming acoustic stimulus ("the seeing ear")[18]. Crucially, empirical perceptual research on error detection uncovers a vital design constraint for software implementation: **error detection accuracy is highest immediately following the first hearing, and systematically degrades on the second and third hearings**[44]. When a user replays an error-filled audio clip multiple times, short-term acoustic memory rapidly encodes and "learns" the incorrect performance, overwriting the user's fragile internal audiation of the written score and causing them to flag correct notes as errors[44]. Consequently, an application error-detection module must be architected around single-pass score comparison, aural change detection, and verbal-label interference[6].

Single-Pass Score-to-Audio Discrepancy Hunt

In the score-based error detection module, the screen displays a prepared score that progresses in complexity across levels from 4-note pitch patterns to 4-measure melodies, two-voice counterpoint, and four-part chorales[44]. Before audio playback is unlocked, the application sounds the tonic reference pitch or cadence and enforces a mandatory 15- to 20-second silent pre-audiation countdown during which the user must read and audiate the written score internally[24]. Once the countdown concludes, the audio plays *once*, containing one or two surgical discrepancies in pitch, scale degree, rhythm, or harmonic inversion[44]. Because repeated hearings degrade internal audiation, replaying the audio before submitting an answer is locked out in assessment mode or incurs a steep score multiplier penalty in practice mode[44]. Rather than merely tapping the measure in which a mistake occurred, the user completes a two-step diagnostic correction: tapping the exact errant notehead on the staff and then selecting or singing the scale degree that was *actually performed* alongside the pitch that was *notated*, directly simulating the diagnostic precision required in ensemble leadership[12].

Aural-to-Aural Change Detection (Sternberg & Same/Different Tasks)

For beginners or non-readers, error detection begins at the purely aural level through Karpinski and Gordon's Same/Different discrimination and Sternberg memory-probe tasks[6]. In the Same/Different variation, the application plays a short melodic, rhythmic, or harmonic pattern followed by a 2-second pause and a second rendition that is either an exact repetition/transposition or contains a single targeted alteration in interval quality, bass

inversion, or metric placement[44]. In the **Sternberg Probe** variant, the application plays a 5-note melody followed immediately by a single isolated probe pitch or solfege syllable, and the user must instantly indicate whether that probe pitch was present in the preceding melody[6]. This simple mechanism trains active listening, rapid short-term encoding, and internal re-hearing without requiring staff literacy[6].

Solfege Stroop Interference ("Mislabeled Syllable" Detection)

Proposed by Chenette to test the resilience of a learner's scale-degree associations, the **Solfege Stroop** exercise introduces deliberate cognitive interference between pitch perception and verbal labels[6]. The application presents a short melody in one of two ways: either displaying a notated melody on screen with one or two *incorrect* solfege syllables printed beneath the notes (for example, printing Do – Re – Fa – So underneath the pitches C – D – E – G), or playing a synthesized vocal track that sings the correct pitches of a melody while pronouncing an incorrect solfege syllable on a target beat[6]. The user must override the verbal distraction, tap the mislabeled note, and sing the entire melody back into the microphone using the *correct* functional solfege syllables[6].

Real-Time DSP Architecture and Vocal Scoring Pipeline

Integrating audiation, sight-singing, tonal pattern echoing, and guide-tone vocalization into an application requires a low-latency, musically intelligent Digital Signal Processing (DSP) pipeline[46]. Naive pitch-detection implementations frequently frustrate users due to mobile microphone gain variance, speaker bleed, octave-locking errors, and harsh penalties applied to natural vocal vibrato or initial onset scoop[48]. Drawing from modern production vocal-tracking architectures, the application's client-side audio engine should implement a multi-stage DSP pipeline that normalizes mobile microphone input, rejects unvoiced consonants, tracks vibrato centers, and scores pitch stability alongside cent deviation[46].

DSP ARCHITECTURE SPEC | Low Latency: 10–25 ms

Real-Time DSP Architecture and **Vibrato-Aware** Vocal Scoring Pipeline

STAGE 1–6 CLIENT-SIDE AUDIO PIPELINE (WEB AUDIO API / NATIVE DSP)

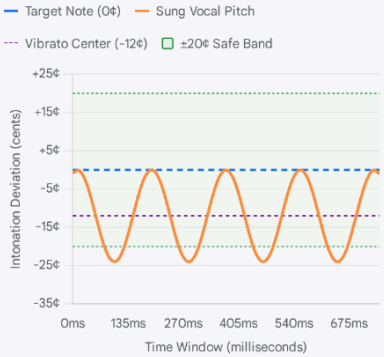
Zero Server Transmission • 100% On-Device

01 • INGEST	02 • FILTER	03 • ESTIMATE
Mic & Auto-Gain	**IIR Bandpass**	**PYIN / FLWT**
44.1 kHz PCM	200 – 3,500 Hz	1024–2048 N
Up to 30x gain if peak < 0.01	Mutes stage bleed & mic rumble	10–25 ms latency window (80–1000 Hz)

04 • GATE	05 • ADAPT	06 • SCORE
CMNDF Check	**Vibrato Center**	**Gaussian Stddev**
Confidence < 0.30	4.5 – 7.0 Hz Mode	15-Frame MIDI
Rejects sibilants, breaths & noise	Upper edge tracking (no false flat)	σ < 0.3 semitone = 85–100% score

Vibrato Trajectory & Pitch Asymmetry

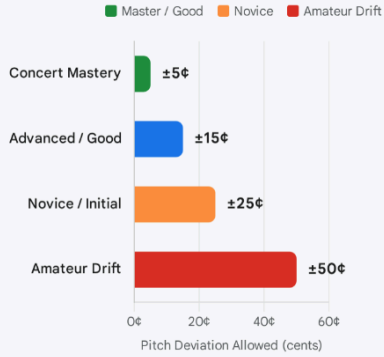
Natural 5.5 Hz downward swing evaluated by Vibrato Center & Upper Edge



Singers naturally swing downward during vibrato. Standard averaging marks held notes flat (-12¢); IntonationPro logic locks to the upper edge and center to reward in-tune audiation.

Intonation Tolerance Tiers

Max allowed deviation in cents before score penalties apply

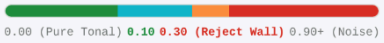


Recommended progression: start beginners at ±20–30 cents tolerance; progress to ±10–20 cents for advanced ear training, down to ±5 cents for concert mastery.

YIN CMNDF Periodicity Gate

Cutoff ≥ 0.30

Normalized difference minimum measuring harmonic tonality vs. unvoiced noise.



`0.00 (Pure Tonal) 0.10 0.30 (Reject Wall) 0.90+ (Noise)`

Clear Singing Voice Periodic vocal tract oscillation	`0.01 – 0.10 CMNDF`
Voiced Vowels & Inflections Evaluated with dynamic EMA weighting	`0.10 – 0.29 CMNDF`
Unvoiced Speech & Room Bleed Consonants, breath, speaker echo muted	`0.30 – 0.90+ (REJECTED)`

Gaussian Pitch Stability Metric

15-Frame Rolling Window

Evaluates pitch-holding stability via standard deviation (σ) of rolling MIDI values.

● **Steady Pitch Lock** Stddev σ < 0.3 semitones	**85% – 100%**	SCORE YIELD
● **Controlled Singing** Stddev σ < 0.5 semitones	**70% – 84%**	SCORE YIELD
● **Unstable Wobble / Drift** Stddev σ > 2.0 semitones	**~0%**	PENALIZED

To evaluate vocal audiation exercises without false penalties, the client-side audio engine processes microphone input within a 10–25 ms latency window. Unvoiced sibilants and speaker bleed are rejected via a YIN CMNDF periodicity gate (>= 0.30), while sustained pitches are judged by the vibrato center and upper edge rather than raw downward pitch swings.

Data sources: Google Play (Shruti Classical Pitch Detector), Apple App Store (IntonationPro Vibrato Center Engine), Vocal Pitch Tuneup Lab, Google Play (Shruti Ear Training Specifications), Frank Karaoke Architecture (YIN CMNDF & Gaussian Pitch Scoring), ResearchGate (Real-Time Wavelet Pitch Tracking).

Audio Ingestion, Gain Normalization, and Bleed Filtering

Mobile hardware introduces severe variance in raw PCM audio levels; for instance, with Automatic Gain Control (AGC) disabled to preserve dynamic integrity, many mobile phone microphones produce peak PCM levels of only $0.05–0.15$, compared to $0.50–0.80$ on desktop microphones[48]. When incoming signal peaks fall below 0.01, the ingestion stage must apply dynamic software gain (up to $30\times$) prior to pitch estimation so the signal reaches usable thresholds for autocorrelation[48]. The boosted signal then passes through a $200–3500\ \text{Hz}$ IIR bandpass filter to attenuate low-frequency room rumble and high-frequency instrumental speaker bleed[48]. Furthermore, to prevent the application's own reference cadence or drone from being scored as the user's voice when practicing without wired headphones, the engine caches the reference audio's timestamped pitch timeline (the "pitch oracle") and ignores microphone frames whose pitch class matches the reference playback whenever vocal confidence falls below 0.30 [47].

Low-Latency Pitch Estimation and Periodicity Gating

To provide instantaneous visual feedback without disorienting lag, the pitch tracker should operate at a $44.1\ \text{kHz}$ sampling rate using 1024- or 2048-sample analysis windows, which correspond to minimum pitch-determination latencies of approximately $25\ \text{ms}$ and $50\ \text{ms}$ via Fast Lifting Wavelet Transform (FLWT), or $10–15\ \text{ms}$ using the probabilistic **PYIN** algorithm across the $80–1000\ \text{Hz}$ vocal range[47].

To prevent unvoiced consonants, breath noise, and sibilants from triggering erratic visual spikes or false pitch penalties during solfege singing (such as the initial consonants in So, Fa, or Ti), the engine monitors the YIN **Cumulative Mean Normalized Difference Function (CMNDF)** minimum value, which directly measures signal periodicity[46]:

$$\text{CMNDF Gate}: \begin{cases} 0.01 \le d'(\tau) \le 0.10 & \text{Clear Tonal Singing (Valid Frame)} \\ 0.10 < d'(\tau) < 0.30 & \text{Transitional / Soft Phonation (Valid if Amplitude } > \theta) \\ 0.30 \le d'(\tau) \le 0.90 & \text{Unvoiced Sibilants, Speech, or Noise (Reject Frame)} \end{cases}$$

Combined with an 85% acoustic machine-learning confidence threshold, gating out frames where $\text{CMNDF} \ge 0.30$ ensures clean note-onset and offset boundaries[46].

Vibrato-Center Judgment and Multi-Metric Scoring

Evaluating human singing purely via raw cent deviation from Equal Temperament—or 12-note Just Intonation when practicing over a static drone—creates two significant pedagogical hazards relating to natural vocal vibrato and autocorrelation snapping[47]. First, natural expressive vocal vibrato modulates at $4.5–7\ \text{Hz}$, and acoustic analysis demonstrates that singers frequently swing *downward* from the target pitch during vibrato cycles rather than

oscillating symmetrically around the mathematical center[46]. Because a naive pitch tracker marks the lower trough of this vibrato as flat, the application's "Singer Mode" algorithm must evaluate held notes, intonation drift, and settling time by tracking both the **vibrato center and upper edge**, protecting intentional $4.5–7\ \text{Hz}$ periodicity while penalizing only persistent flat or sharp drift[46].

Second, because standard YIN autocorrelation naturally snaps toward harmonic frequencies near semitone boundaries, raw cent deviation can prove less discriminative than expected during rapid melodic movement[48]. Therefore, the scoring engine should pair **Configurable Cents Tolerance Bands**—starting beginners at a generous $\pm 20 \text{ to } 30+\ \text{cents}$ tolerance, tightening to $\pm 10–20\ \text{cents}$ for intermediate singers, and offering a $\pm 5\ \text{cents}$ mastery setting—with a **Pitch Stability** metric[47]. Pitch Stability calculates a Gaussian decay score based on the standard deviation (σ) of detected MIDI pitch values across a rolling 15-frame ($\sim 450\ \text{ms}$) window[48]. Steady vocal internalization ($\sigma <$) yields $85–100\%$ credit, acceptable stability ($\sigma <$) passes, and pitch hunting or vocal wobbling ($\sigma > 2.0\ \text{semitones}$) scores near 0%[48]. Additionally, if a user's average pitch marker sits consistently sharp or flat across an entire song or session, the application can diagnose a constant reference-tuning offset rather than a relative-pitch error, offering targeted vocal-motor calibration exercises[21].

Implementation Roadmap and Curricular Synthesis

To integrate these exercises cohesively without overwhelming the user interface, the application should organize its relative pitch and audiation curriculum into a four-tier developmental progression, pairing adaptive error tracking—such as a confusion matrix that identifies specific scale-degree pairs the user conflates, like La heard as Sol, and dynamically increases their presentation frequency—with structured daily 10-minute practice sessions[4].

Curriculum Tier	Core Cognitive Objective	Integrated Application Modules & Exercise Mechanics	Target Mastery & DSP Thresholds
Tier 1: Tonal Context & Internal Anchor	Establish resting-tone stability (Do/La), vocal-motor pitch matching, and basic scale-degree	Resting Tone Drop-Needle (Do/La mid-phrase halt); Upgraded Scale Degrees (Benbassat ladder	90% accuracy across 20-question sessions; Vocal tolerance $\pm 25–30\ \text{cents}$; Stability

	gravity[4].	with 1-beat breath pause and post-answer stepwise tonic resolution trace); Aural Same/Different & Sternberg Probe change detection[4].	$\sigma <$ [4].
Tier 2: Tonal Vocabulary & Working Memory	Build a 2–5 note functional pattern vocabulary (Tonic/Dominant) and expand short-term pitch memory[6].	MLT Tonal Pattern Echo (Neutral "bum" $\rightarrow$ Solfege Do-Mi-So / So-Fa-Re-Ti with Teaching/Evaluation passes); Functional I–V Dialogue (improvise contrasting function patterns & isolate starting pitches); Musical N-Back (1-Back & 2-Back recognition & vocal canon); Additive & Complementary 2-step interval scaffolding[6].	Mandatory 1-beat inhalation delay before mic opens; Vocal tolerance $\pm 15–20$ cents; Vibrato center and upper-edge tracking active[13].
Tier 3: Inner Hearing & Linear Harmony	Maintain internal pitch across silence, manipulate mental pitch representations, and track linear voice-leading[2].	Kodály Audiation Gate (sight-singing with muted bars and strong/weak beat inner-hearing drop-outs); Melodic Word-Updating (silent mental replacement of target scale degrees); Bass &	Re-entry pitch on Beat 1 after 2-bar silence within ± 15 cents; Stability $\sigma <$ (85–100% score band)[2].

		Guide-Tone Vocalizer (sing chord roots and 3rd/7th lines over static tonic reference and rootless voicings); Non-Chord Tone Skeleton Explorer (toggle skeleton vs. PT, NT, SUS, APP, ANT, ET)[2].	
Tier 4: Structural Synthesis & Error Detection	Translate complex polyphonic/modulating sound to symbol and detect real-time performance discrepancies[18].	Extractive Listening & Protonotation Dictation (selective voice memorization and pulse/contour sketching prior to staff notation); Single-Pass Score Error Detection (15s silent score audiation followed by single-play discrepancy hunt); Solfege Stroop Mismatch (override mislabeled syllables); Modulation & Cadence Tracker (resolve secondary dominant tendency tones Fi $\rightarrow$ So to new resting tones)[6].	Single-hearing error detection accuracy $\geq$; Optional $\pm 5–10$ cents mastery tuning in Equal or 12-note Just Intonation[44].

By shifting the software's pedagogical center of gravity away from isolated interval arithmetic and toward **resting-tone inference**, **Gordon tonal-pattern conversations**, **Kodály silent-singing gates**, **working-memory pitch manipulation**, **linear guide-tone resolution**, and **single-pass error detection**, the application will directly train the cognitive mechanisms

that define elite relative pitch and musical audiation[2].

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